Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

For layered thin films, the central structural question is often not only which phase is present, but how the crystallites are oriented relative to the substrate. In layered van der Waals solids and related two-dimensional materials, this orientational state can directly control the properties that motivate the films in the first place.[1–3] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[4] In topological bismuth chalcogenides, strong c-axis alignment is needed to preserve the desired electronic response.[5] In transition-metal dichalcogenides, catalytic activity can increase when platelet edges are exposed by the film orientation.[6] Comparable orientational states also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films such as $Ti_3C_2T_x$ MXenes and graphene-oxide papers.[7–14]

Wide-angle x-ray scattering (WAXS) with an area detector is attractive for this problem because a single exposure probes a large reciprocal-space volume rich with information of both structure and orientation.[15–19] The difficulty is that many layered films do not fall cleanly into either of the standard diffraction limits. They are neither single crystals, where reciprocal space consists of discrete points, nor fully isotropic three-dimensional (3D) powders, where orientational averaging produces spherical shells.[20] Instead, a common thin-film orientational state is the two-dimensional (2D) oriented powder, in which plate-like crystallites share an aligned layer normal while remaining azimuthally randomized in-plane.[3, 20, 21] This intermediate orientational state produces detector patterns that inherit features of both limiting cases while necessitating nontraditional analysis techniques.[20]

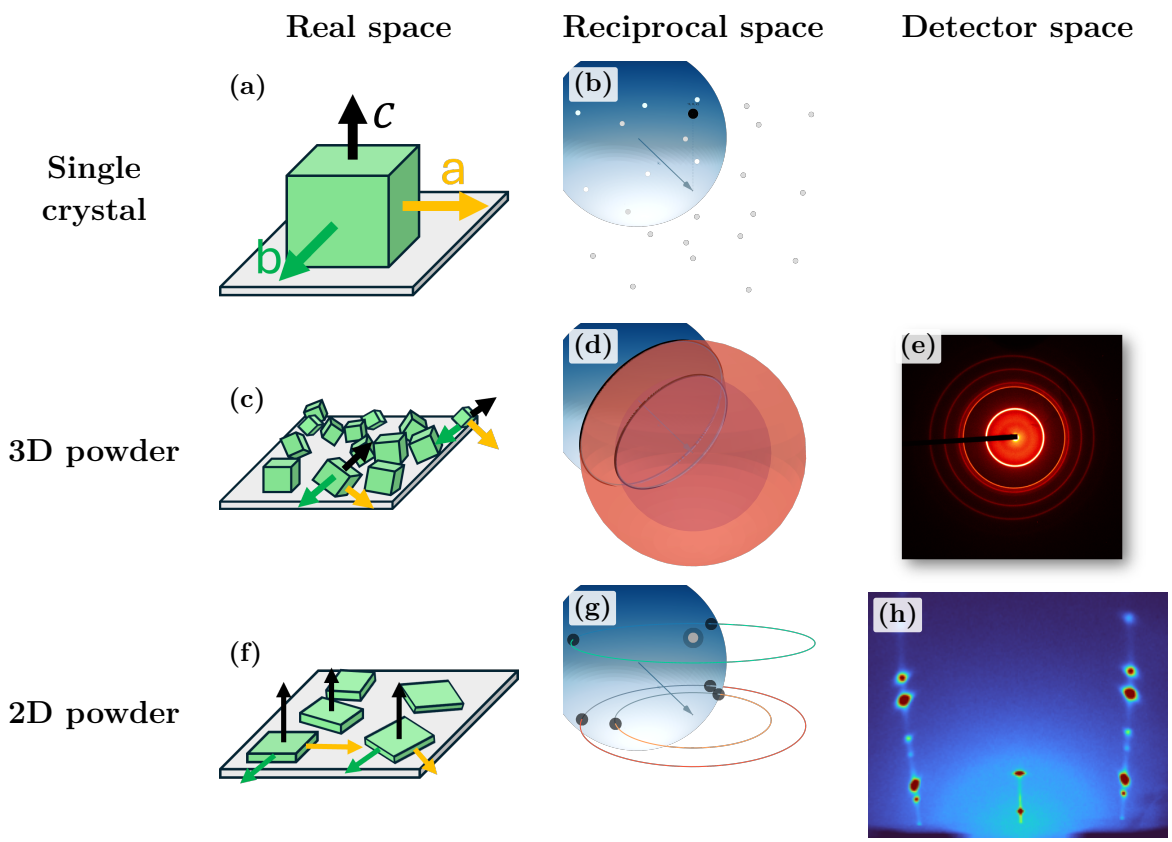

Figure 1: Diffraction across three orientational limits: single crystal, 3D powder, and 2D powder. Columns show the corresponding real-space, reciprocal-space, and detector-space views, while rows group the orientational limits. Panels (a,b) show a single crystal in real space and reciprocal space. Panels (c–e) show a 3D powder, with randomly oriented crystallites, spherical reciprocal-space shells, and Debye–Scherrer rings on the detector. Panels (f–h) show a 2D powder, with aligned surface normals and in-plane orientational disorder, reciprocal-space rings with specular Bragg features, and the corresponding detector pattern with symmetry about the central axis.

Figure 1 summarizes the real-space, reciprocal-space, and detector-space consequences of these three orientational limits. In the Ewald construction, elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[22] For a single crystal, reciprocal space consists of isolated Bragg points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically intersects only a small subset of reflections and the detector records sparse spots.[22] For a 3D powder, random crystallite orientations smear the intensity of each Bragg point over a spherical shell, and the Ewald-sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1d, Fig. 1e).[22] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1g).[15, 20] The corresponding illustration in Fig. 1f schematically shows 2D platelets with aligned growth, and a representative 2D-powder detector pattern is shown in Fig. 1h. When the Ewald sphere intersects this reciprocal-space distribution, the detector records hybrid signatures: pairs of symmetric spots from the off-specular ring intersections and single-crystal-like features near

the specular stripe from $(00L)$ intensity.[15, 17, 23] Mosaicity then redistributes intensity, broadening the ring intersections into arcs and spreading the $(00L)$ features into cap-like intensity near the specular region.[15, 20]

This hybrid detector-space information is what makes the 2D-powder problem both rich and difficult. Many standard WAXS workflows reduce area-detector images to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[17, 19, 24–26] Those steps suit nearly isotropic powders, but for 2D powders they deweight or remove off-specular arcs and specular caps that constrain out-of-plane correlations, stacking, and mosaic tails.[15, 17] Stacking disorder is also commonly inferred from 1D interference models and refinements,[27–29] which do not directly exploit the two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with orientation-distribution-function (ODF) refinements or preferred-orientation corrections,[30–35] but those approaches are more general than needed for the narrowly constrained 2D-powder case considered here.[34, 35] Recent radial-profile approaches have made thin-film GIXD substantially more quantitative, but peak overlap and background separation can remain limiting in strongly anisotropic patterns, especially at small incidence angles.[36] For the layered films considered here, the measured morphology already motivates axial symmetry about the film normal and a low-dimensional out-of-plane mosaicity kernel.[3, 20, 21] We therefore assume this physically motivated 2D-powder orientation distribution a priori and refine it with a detector-space forward model, using off-specular arcs and specular-family line shapes as direct constraints.[15, 17]

Here we present a detector-space forward model for 2D-powder WAXS. Starting from an indexed crystallographic model, it predicts area-detector images and applies the same projection and integration procedures to measured and calculated intensity. The comparison uses detector-space peak positions, local line shapes, relative Bragg intensities, and projected Q_z profiles. Prior work has established the main ingredients needed for such a framework, including grazing-incidence scattering formalisms,[37, 38] reciprocal-space mapping and indexing for oriented thin films,[15, 17, 19, 23] and forward calculations of diffraction from textured films.[39] Complementary atomistic tools have also been developed for constructing commensurate multilayer 2D heterostructure supercells and visualizing reciprocal-space diffraction signatures, including Nookiin.[40] In contrast, the present work carries the calculation in detector space through detector geometry, grazing-incidence propagation, beam phase space, mosaicity, resolution, and ordered structure-factor weighting. The main-text validation is centered on the ordered 2D-powder limit using Bi_2Se_3 and Bi_2Te_3 films.[5, 7] Diffuse PbI_2 and stacking-disorder refinements are treated as later extensions of the same framework rather than as the primary proof of the method.

2 Diffraction Model

The detector-space forward model redistributes the structure-factor intensity of the contributing reflections over mosaic distributions. Monte Carlo samples the incident beam profile across those orientations, weights each ray by transmission, and maps the summed intensity to detector pixels. The calculated and measured images are then compared using the same peak positions, line shapes, relative Bragg intensities, and projected Q_z profiles used for the ordered-film validation.

2.1 Diffraction Geometry

We first consider a single incident ray striking a perfectly aligned crystallite, with no beam divergence, no refraction, and no absorption. Elastic scattering requires

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}, \qquad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i.$$

Diffraction occurs when the momentum transfer matches a reciprocal-lattice vector, $\mathbf{Q} = \mathbf{G}$. In the Ewald construction, this means that the reciprocal-space point for that reflection lies on the Ewald sphere of radius $|\mathbf{k}_i|$, and the simplest diffracted ray is obtained from

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}.$$

For a fixed $|\mathbf{G}|$, the set of possible reflection orientations forms a Bragg sphere of radius $|\mathbf{G}|$ about the reciprocal-space origin. The allowed exit vectors are found by intersecting that Bragg sphere with the Ewald sphere and then propagating the resulting exit ray to the detector.

For each trial incident ray, the model computes the sample intersection, applies the incidence geometry and optical corrections, finds the reciprocal-space intersections that satisfy elastic scattering, and propagates the resulting exit rays to the detector. Rays are Monte Carlo sampled from the probability distribution formed by the shape, bandwidth, and divergence of the beam. A second Monte Carlo sampling occurs for each ray to choose an intersection point. Footprint, detector tilt, refraction, and absorption enter as experimental resolution and intensity terms. The coordinate conventions for the lab frame, detector plane, sample rotation, and illuminated footprint are summarized in Figs. 2 and 3.

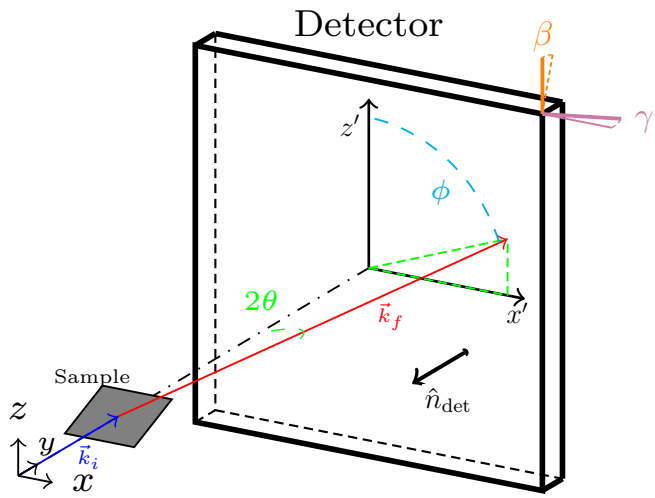

Figure 2: Overall lab-frame geometry. The incident wavevector $\mathbf{k}_i$ strikes the sample, and a representative exit wavevector $\mathbf{k}_f$ reaches the detector at scattering angle 2θ and detector-plane azimuth ϕ. The detector at distance D is tilted by β and γ.

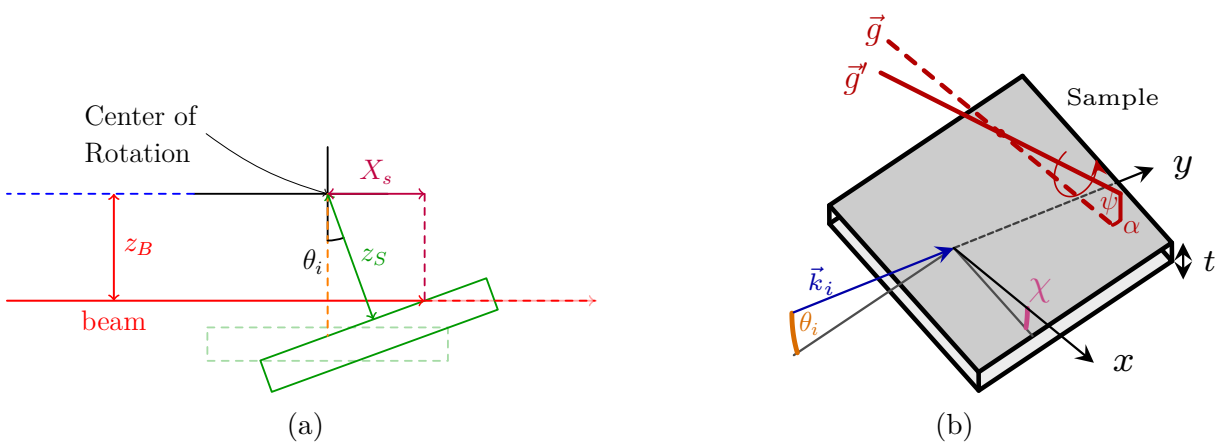

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

We enumerate integer in-plane Miller pairs (H, K), construct the reciprocal-space rod along G_z for each pair, and evaluate its detector intersections. We define HK as the scalar in-plane metric sum

$$\mathrm{HK}(H, K) = \begin{bmatrix} H & K \end{bmatrix} \mathbf{M}_{\parallel} \begin{bmatrix} H \\ K \end{bmatrix} \propto Q_r^2,$$

where $\mathbf{M}_{\parallel}$ is the in-plane reciprocal-lattice metric, with any lattice-dependent normalization absorbed into the definition of HK. Thus, HK labels an in-plane reciprocal-space rod family. In 2D-powder calculations, all integer (H, K) rods with the same HK, and therefore the same Q_r, are evaluated individually and then summed after projection. Multiplicity and orientation effects therefore emerge within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a \, f_a(Q) \, \exp[2\pi i \, \mathbf{G} \cdot \mathbf{r}_a] \, \exp\Big[-2\pi^2 \, \mathbf{G} \cdot \mathbf{U}_a \cdot \mathbf{G}\Big] \,, \tag{1}$$

where the sum runs over atoms a in the unit cell. Here o_a, $f_a(Q)$, $\mathbf{r}_a$, and $\mathbf{U}_a$ are the occupancy, atomic form factor, fractional position, and anisotropic displacement tensor for atom a. Symmetry constraints are retained during refinement, so only the independent crystallographic parameters are varied. The detector weight for each allowed ray is multiplied by $|F_{\mathbf{G}}|^2$ before mosaic and experimental broadening are applied.

2.3 Mosaicity

Mosaicity can be understood by comparing the orientational limits introduced in Fig. 1. For a perfect single crystal, a reflection carries its full weight at a single Bragg spot. For a 3D powder, random crystallite orientations spread that weight isotropically over the Bragg sphere. A 2D powder lies between these limits: in-plane rotations are randomized about the film normal, while the layer normals retain a finite out-of-plane misorientation distribution. Mosaicity is therefore treated as a redistribution of the ideal per-reflection intensity over the allowed HK family.

For a reflection, we work in the (G_r, G_z) cross-section of reciprocal space, where G_r is the in-plane radial component. Random in-plane rotations generate a ring of equivalent reciprocal-space intersections about G_z. Out-of-plane misorientation then gives a nonuniform line density around that ring, written as $I_0\omega(\Delta\theta)$, where I_0 is the reflection intensity before mosaic redistribution and $\Delta\theta$ is the crystallite tilt relative to the mean layer normal.

The ordered-film images show why a single narrow mosaic distribution is insufficient. At a fixed nominal incident angle θ_i, the measured $HK = 0$ specular family contains $00L$ peaks that are not directly selected by the ideal incidence condition. A narrow Gaussian-like core can reproduce the main widths of near-condition off-specular arcs, but it decays too rapidly to supply enough diffracting crystallites for weak off-condition specular intensity. The 003 detector feature isolated in Fig. 5 is one visible example of this problem: the observed intensity requires crystallites whose layer normals are tilted far enough from the mean orientation to satisfy the Bragg condition, together with the finite wavelength bandwidth of the incident

beam.

We therefore describe the out-of-plane mosaicity by a two-component angular density,

$$\omega_{\mathrm{w}}(\Delta\theta;\eta,\Gamma,\sigma) = \eta L_{\mathrm{w}}(\Delta\theta;\Gamma) + (1-\eta)G_{\mathrm{w}}(\Delta\theta;\sigma), \qquad 0 \le \eta \le 1. \tag{2}$$

Here G_{w} is the wrapped Gaussian core, L_{w} is the wrapped Lorentzian tail, and η is the Lorentzian weight. The wrapped form is evaluated on $(-\pi,\pi]$ so that probability is conserved over one full angular period and the distribution has no artificial discontinuity at $\Delta\theta = \pm\pi$. The widths are independent: the Lorentzian full width at half maximum is 2Γ, while the Gaussian full width at half maximum is $2\sqrt{2\ln 2}\,\sigma$. For $|\Delta\theta| \gg \Gamma$, the Lorentzian component decays algebraically, $L(\Delta\theta;\Gamma) \simeq \Gamma/(\pi\Delta\theta^2)$, rather than exponentially. This long tail supplies the small but observable tilted-crystallite population needed for off-condition $00L$ peaks. The full periodic forms of L_{w} and G_{w} are given in the supporting information.

Figure 4 summarizes how this reciprocal-space broadening appears in the $\{G_r, G_z\}$ plane and how it maps onto characteristic detector-space features for in-plane and specular conditions.

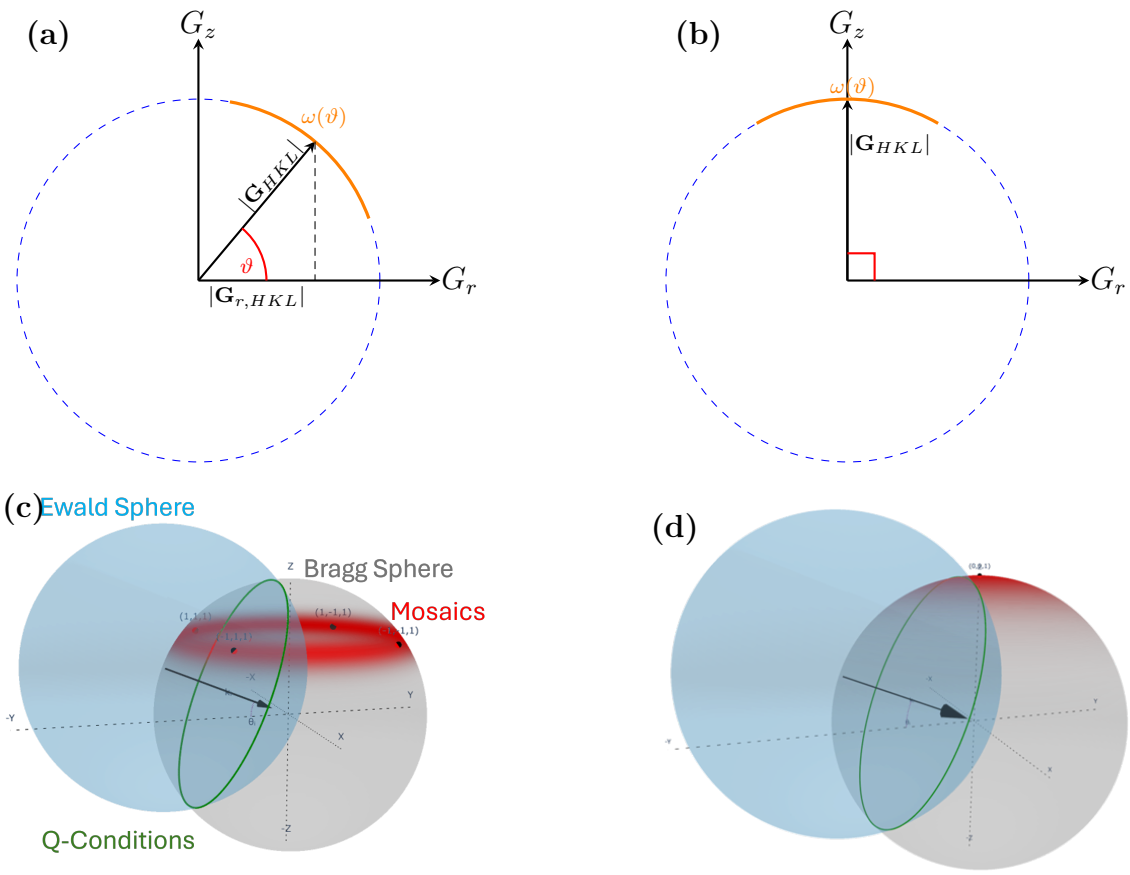

Figure 4: Mosaic broadening viewed in the $\{G_r, G_z\}$ plane (a,b) and its detector-space manifestation (c,d). Intensity is redistributed along the Bragg circle with line density $I_0\omega(\Delta\theta)$. The in-plane condition with $G_r \neq 0$ (a,c) yields two off-specular arcs, while a specular reflection with $G_r = 0$ (b,d) concentrates intensity near the polar cap. The long tail of $\omega(\Delta\theta)$ is what gives finite intensity to $HK = 0$ specular-family peaks that are not directly excited by the nominal incidence angle.

After random in-plane rotation, the one-dimensional tilt density becomes an axially symmetric surface density on the Bragg sphere. Let ϑ be the polar angle on a Bragg sphere of radius $G \equiv |\mathbf{G}|$, and let ϑ_B be the Bragg position for the mean crystallite orientation, so that $\Delta\theta = \vartheta - \vartheta_B$. Conservation of the reflection intensity over the spherical area element $dA = 2\pi G^2 \sin\vartheta \, d\vartheta$ gives

$$\rho(\vartheta) = \frac{I_0}{2\pi G^2 \sin\vartheta} \, \omega_{\mathrm{w}}(\vartheta - \vartheta_B), \qquad 0 < \vartheta < \pi. \tag{3}$$

The $1/\sin\vartheta$ factor is the geometric Jacobian for revolving the one-dimensional tilt distribution into a Bragg-sphere surface density. This density supplies the weights used to distribute I_0 over mosaic-broadened exit directions; the two-component form in Eq. (2) determines whether those weights vanish rapidly away from the nominal condition or retain the long tail needed for the observed $HK = 0$ intensity.

3 Refinement workflow

We use a staged refinement because geometry, mosaicity, and crystallographic intensity are strongly correlated in a fully global fit. Each stage uses the detector-space observable most sensitive to one parameter group before the next group is introduced. Table 1 summarizes the ordered-film workflow used for Bi_2Se_3 and Bi_2Te_3.

Table 1: Staged refinement workflow used for the ordered Bi_2Se_3 and Bi_2Te_3 films. All sample images are dark-subtracted before the listed observables are formed. Diffuse-scattering and stacking-disorder refinements are deferred from the present main-text draft.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x$, $\Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D, detector tilts (β, γ), and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry $(\theta_i, \delta, z_B, z_S)$, in-plane rotation χ, goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected local detector ROIs, specular-tail checks, and local profile shapes	Gaussian-plus-Lorentzian mosaic parameters	Center-aligned profile shapes, off-specular arcs, and specular tails
5	Ordered-film validation	Selected detector-space Bragg regions and Q_z projections from fixed-Q_r/HK trajectories	Image scale factors and ordered structure-factor parameters	Relative Bragg intensities and measured/calculated line-shape agreement

The instrumental and alignment stages fix where an ideal reflection should appear on the detector. Direct-beam and calibrant data define the beam phase space and detector mapping, while indexed sample reflections determine the incidence condition, sample offsets, and goniometer misalignment.

The remaining stages refine peak shape and intensity. Local detector profiles constrain the shared Gaussian-plus-Lorentzian mosaic distribution. Selected Bragg-region integrals constrain ordered structure-factor parameters, and the final validation projects measured and calculated images along the same fixed-Q_r/HK trajectories for comparison versus Q_z. Implementation details required for efficiency, including 2θ–ϕ remapping, caked fields, lookup tables, and Monte Carlo sampling, are described in the supporting information. Later stacking-disorder will be acquired by fitting the HK rods as a function of L.

4 Results: ordered Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films validate the model by showing that the same detector-space forward model quantitatively reproduces their measured line shapes and intensities. In the fitting process, three detector images with incident angles of 5°, 10°, and 15° were used. Together, they test the 2D-powder orientational limit, with sample geometry, beam phase space, mosaicity, detector resolution, and ordered structure-factor weighting. Both

materials exhibit the expected 2D-powder signatures: off-specular Bragg arcs from in-plane HK families and specular-family intensity from $HK = 0$ reflections.

For each material, the comparison is shown in two complementary views. The detector-space panel identifies the fixed-HK Bragg-rod trajectories used for projection while keeping the measured area-detector intensity visible. The profile panel compares measured and calculated intensity plotted on the same axes as a function of L for the selected HK families. The detector image is the map of where the profiles are extracted. Each HK label denotes a family of symmetry-equivalent rods with the same Q_r. Those rods are generated and projected separately before their detector-space contributions are summed. The plus and minus branches, where shown in profile panels, refer only to the two detector-side intersections of the same in-plane rod family and are not separate crystallographic families.

A central result of the ordered-film comparison is the need for a Lorentzian contribution to the mosaic distribution. The clearest evidence is the $HK = 0$ specular family, i.e. the $00L$ peaks. For a perfectly aligned crystallite at the nominal incidence angle θ_i, only the specular condition directly selected by that incidence geometry should carry appreciable intensity. The measured detector images and projected profiles instead contain additional $HK = 0$ peaks whose Bragg conditions are not directly excited by θ_i. A narrow Gaussian-like mosaic core can account for the strong near-condition arc widths, but it decays too rapidly to supply enough diffracting volume for these off-condition specular peaks. The Lorentzian tail represents crystallites tilted far enough from the mean film normal to bring those reflections onto the Ewald sphere.

A second important feature appears near the 003 condition, or $HK = 0$, $L = 3$, where the measured image shows a star-like pattern together with bright intensity extending above the critical-angle region (Fig. 5). This is the observational problem that motivates the long-tail mosaic component before the full Bi_2Se_3/Bi_2Te_3 comparison is shown. A narrow mosaic core would keep most crystallites close to the nominal film normal and would strongly suppress intensity from off-condition $00L$ peaks. The measured 003 feature instead requires a population of crystallites tilted far enough from the mean orientation to satisfy the Bragg condition, together with the finite bandwidth of the incident beam.

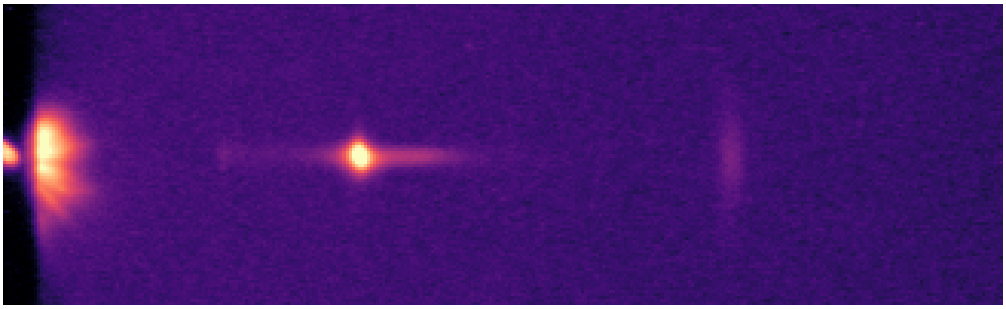

Figure 5: Close-up of the 003 detector feature, equivalently $HK = 0$, $L = 3$. The star-like intensity above the critical-angle region is not assigned to reflectivity. Instead, the finite incident bandwidth gives the Ewald sphere an effective thickness, which intersects the small-radius 003 Bragg sphere over a finite volume. Combined with the long mosaic tail, that overlap redistributes intensity into the observed detector feature and motivates including the Lorentzian mosaic component before evaluating the full data/model overlays.

This feature is therefore explained within the same diffraction picture as the other $00L$ peaks rather than as an unrelated detector artifact. The finite incident bandwidth gives the Ewald sphere an effective thickness, while the Lorentzian mosaic tail supplies crystallites with enough out-of-plane misorientation to bring the 003 Bragg sphere into the accepted scattering condition. The ordered-film validation below then tests whether this same phase-space treatment also reproduces the full set of off-specular and specular line shapes.

Figure 6 gathers the ordered-film validation into a single matched comparison. Panels (a) and (b) are the detector-space views for Bi_2Se_3 and Bi_2Te_3, respectively. These panels identify the fixed-Q_r/HK trajectories used for projection and show both off-specular and $HK = 0$ features in the measured images. Panels (c) and (d) show the corresponding projected Q_z profiles for Bi_2Se_3 and Bi_2Te_3. In each profile panel, the measured and calculated intensities have been subjected to the same projection and integration procedure.

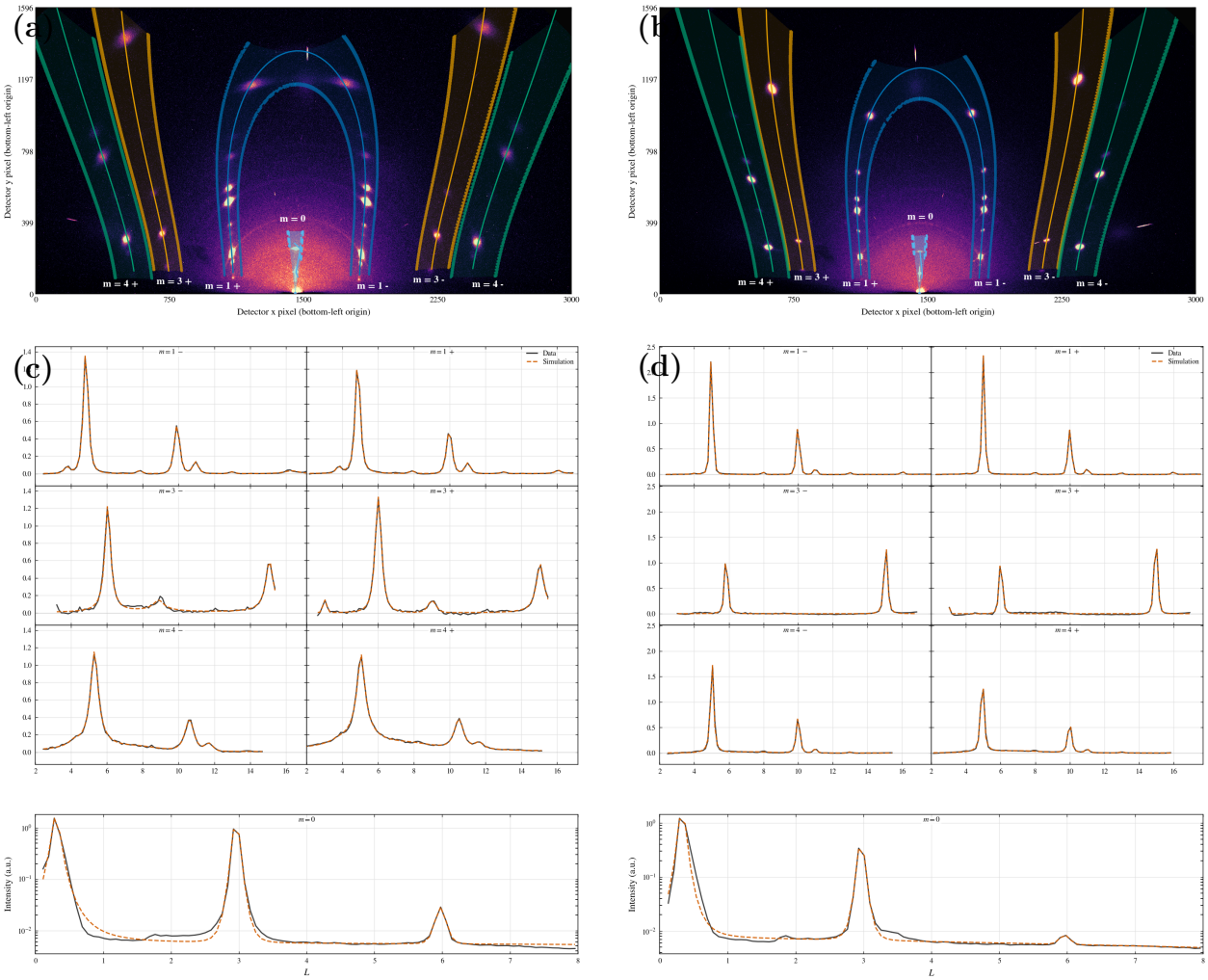

Figure 6: Model and data comparison for Bi_2Se_3 and Bi_2Te_3. (a,b) Detector-space views show the selected HK trajectories. (c,d) The corresponding L profiles compare measured and calculated intensities using the same projection and integration procedure. The $HK = 0$ rows probe off-condition $00L$ intensity associated with the Lorentzian mosaic tail.

Because mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, detector resolution, and finite integration width all broaden the effective scattering condition, the projected coordinate should not be interpreted as an ideal reciprocal-space cut from a single perfect crystallite. The comparison remains meaningful because the calculated image is subjected to the same projection and resolution effects before it is compared with the measured profile. The figure therefore tests the detector-space calculation as used experimentally.

In Fig. 6, the off-specular HK and $HK = 0$ profiles play complementary roles. The off-specular arcs check the 2D-powder reciprocal-space construction and calibrated detector geometry, while their widths and asymmetries constrain the Gaussian-like mosaic core and instrumental resolution. The off-condition $HK = 0$ peaks test the Lorentzian tail and show why a rapidly decaying mosaic core alone is insufficient. The bandwidth-sensitive

003 morphology isolated in Fig. 5 reinforces that the forward model must include finite wavelength bandwidth explicitly.

5 Discussion and conclusion

The ordered Bi_2Se_3 and Bi_2Te_3 films validate the mosaic model and detector-space forward model. A single calculation accounts for detector geometry, finite beam phase space, sample alignment, mosaicity, resolution, and ordered structure-factor weighting, then compares measured and calculated Q_z profiles along the same fixed-Q_r/HK trajectories. The key result is direct line-shape agreement.

Keeping the comparison in detector space is essential because the constraining intensity is distributed among off-specular arcs, cap-like $HK = 0$ features, and weak off-condition specular peaks that a conventional 1D reduction would obscure. Because the calculated image includes the same effects and is projected in the same way as the measurement, the overlay tests the complete experimental model. The off-condition $HK = 0$ peaks also show why a narrow Gaussian-like mosaic core is insufficient and why the Lorentzian tail is required.

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